⚙️ MECHANICS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Simple Machines — Same Work with Less Applied Force

- Simple machine ka purpose work ko "kam" karna nahi, balki same amount of work ko comparatively kam applied force / convenient force-direction ke saath karna hota hai. Pulley, lever, wheel-and-axle aur screw common examples hain. *[Q1]*

2. Sensitive Balance & Inhaled Air — Why Reading Increases

- Sensitive balance par khada vyakti deep breath leta hai to inhaled air uske system ke mass mein temporarily add hoti hai, isliye balance reading thodi badh sakti hai. *[Q2]*
- Source explanation ke hisaab se air density roughly 1.225 g/L li gayi hai; several litres ki deep inhalation se few grams ka mass difference possible hai. *[Q2]*

3. Scalar vs Vector Quantities — Speed, Velocity, Force, Pressure & More

- Scalar quantity mein sirf magnitude hota hai; vector quantity mein magnitude ke saath direction bhi hoti hai. *[Q3, Q7]*
- Displacement, velocity, acceleration, momentum, force aur torque vector examples hain. *[Q3, Q4, Q5, Q7]*
- Speed, distance, time, volume, pressure, energy, work aur density scalar examples hain. *[Q3, Q4, Q5, Q6, Q7]*
- Pressure scalar hai, although pressure se associated force direction-specific ho sakta hai. Pressure = Force / Area. *[Q6]*

4. Momentum, Mass & Force — p = mv and F = ma

- Linear momentum $p = m \times v$. Isliye momentum / velocity = mass. *[Q8]*
- Newton's second-law form mein force $F = m \times a$; force mass aur acceleration ka product hai. *[Q9]*
- Non-contact forces bina direct physical contact ke act karte hain: gravitational, magnetic aur electric forces. Friction aur applied/impact force contact forces hain. *[Q10]*

5. Cube Geometry Quick Numerical — Volume = Surface Area

- Cube ke liye volume = l^3 aur total surface area = $6l^2$. Agar numerical values equal hon: $l^3 = 6l^2 \Rightarrow l = 6$ units. *[Q11]*

6. Energy — Conservation, Sources, Wind Power & Energy Conversion

- Law of Conservation of Energy: energy na create ki ja sakti hai, na destroy; sirf ek form se doosri form mein transform/transfer hoti hai. *[Q12]*

- Wind energy moving air ki kinetic energy hai. Wind turbine is kinetic energy ko mechanical energy aur generator usse electrical energy mein convert karta hai. *[Q14, Q15]*
- Friction ke against motion mein mechanical/kinetic energy ka ek part heat mein convert hota hai. *[Q16]*
- Energy-conversion matching: Light → Electric = Solar cell; Electric → Sound = Loudspeaker; Mass → Heat/Energy = Nuclear reactor; Chemical → Heat & Light = fuel combustion. *[Q17]*
- Most biological/wind sources ultimately solar origin se linked hain, jabki nuclear aur geothermal energy direct solar-derived sources nahi hain. *[Q13]*

⚠ EXAM TRAP — Q13 — Nuclear vs Geothermal Source Ambiguity

▸ Question poochta hai "NOT ultimately derived from Sun". Scientifically nuclear energy aur geothermal energy dono direct solar-derived nahi hain. *[Q13]*

▸ HTML options mein A = Nuclear, B = Geothermal hai; explanation dono ko correct bolti hai, lekin stored key A marked hai. Exam source ko exactly recognize karein. *[Q13]*

7. Speed — Galileo and the Basic Relation v = d/t

- Exam-source convention Galileo Galilei ko speed ko distance travelled per unit time ke form mein systematically define/measure karne se associate karta hai. *[Q18]*
- Basic relation: speed v = distance d / time t. *[Q18]*

8. Newton's First Law & Inertia — Train and Turning-Bus Examples

- Newton's First Law ko Law of Inertia kaha jata hai: body rest ya uniform straight-line motion ki state maintain karti hai jab tak unbalanced external force act na kare. *[Q19]*
- Train suddenly start kare to lower body train ke saath move karti hai, upper body inertia of rest ke karan peeche rehne ki tendency rakhti hai; passenger backward lean karta hai. *[Q20]*
- Bus suddenly right turn le to passenger ka upper body inertia of direction ke karan left side ki taraf lean/throw hota dikhta hai. *[Q21]*

9. Statics & Conservation of Momentum — Rest, Whistling and Boat Recoil

- Statics mechanics ki branch hai jo bodies in equilibrium/rest ko study karti hai. *[Q22]*
- Frictionless surface par man air ko expel karke recoil/action–reaction se khud ko propel kar sakta hai; source assertion–reason key dono statements ko true but R ko not-correct-explanation mark karti hai. *[Q23]*
- Boat-man system initially rest mein ho to total momentum zero hota hai. Man 5 m/s se jump kare aur boat 0.5 m/s opposite move kare: $M\times5 = N\times0.5 \Rightarrow$ boat mass = 10 × man mass. *[Q30]*

⚠ EXAM TRAP — Q23 — Assertion–Reason Interpretation

▸ Source key: A and R both true, but R is not the correct explanation of A. *[Q23]*

▸ Conceptually recoil ko Newton's third law aur momentum conservation dono viewpoints se discuss kiya ja sakta hai; exam mein source key follow karna zaroori hai. *[Q23]*

10. Friction — Ice, Loaded Cart, Ball Bearings and Rolling Ball

- Ice par road ke comparison mein friction kam hota hai, isliye walking difficult hoti hai; forward push ke liye sufficient static friction chahiye. *[Q24]*
- Loaded cart ko start karna harder hota hai because limiting/static friction generally kinetic friction se greater hota hai; once moving, required force kam ho jata hai. *[Q25]*
- Ball bearings wheel/axle ke beech sliding contact ko rolling contact mein convert karke friction reduce karte hain; isliye cycles/cars mein use hote hain. *[Q34, Q35]*
- Level ground par rolling cricket ball eventually frictional/resistive forces ke karan slow hokar stop hoti hai; ideal no-friction case mein motion continue karega. *[Q39]*

⚠ **EXAM TRAP — Q35 — Ball Bearing Friction Wording**

▸ Standard physics wording: ball bearings SLIDING friction ko ROLLING friction mein convert/reduce karte hain. *[Q34, Q35]*

▸ Source Q35 mein "static friction → rolling friction" wording diya hai aur rolling friction answer marked hai; wording ko exam trap ke roop mein yaad rakhein. *[Q35]*

11. Newton's Second & Third Laws — Goalkeeper and Action–Reaction

- Goalkeeper fast ball catch karte waqt hands backward pull karta hai, stopping time Δt badhta hai; same momentum change ke liye average force $F = \Delta p/\Delta t$ kam ho jata hai. *[Q26]*
- Newton's Third Law: every action ke liye equal magnitude ka opposite-direction reaction force hota hai; pair different interacting bodies par act karta hai. *[Q27]*

12. Circular Motion & Rotating Frames — Centripetal and Coriolis Forces

- Circular motion ke liye centre ki taraf net force required hota hai, jise centripetal force kehte hain. *[Q28]*
- Coriolis force rotating reference frame ka apparent/inertial force hai. Earth rotation ke karan moving bodies Northern Hemisphere mein right aur Southern Hemisphere mein left deflect karti hain. *[Q29]*
- High-speed sharp turn par insufficient banking/friction ki condition mein vehicle outward skid kar sakta hai aur outer wheels ke about overturn tendency develop ho sakti hai. *[Q38]*

13. de Broglie & Work–Energy Numericals — Wavelength and Kinetic Energy

- de Broglie relation $\lambda = h/p = h/(mv)$. For $m = 0.12$ kg, $v = 20$ m/s, source calculation $\lambda \approx 2.76 \times 10^{-34}$ m deta hai. *[Q31]*
- Work–Energy theorem: stopping work $F\times s$ = initial kinetic energy. Equal KE aur equal stopping force ho to truck, car aur motorcycle equal stopping distance cover karenge: $X = Y = Z$. *[Q32]*
- Same mass ke liye $KE \propto v^2$. Agar $K_1:K_2 = 4:9$, to $v_1:v_2 = \sqrt{4}:\sqrt{9} = 2:3$. *[Q33]*

14. Vehicle Dynamics in Accelerating Frames — Car Wheels and Oil Tanker

- Accelerating car mein inertia/pseudo-force ke effect se load front se rear ki taraf transfer hota hai; acceleration ke during front-wheel normal reaction R se kam ho sakti hai. *[Q36]*
- Forward uniformly accelerating oil tanker mein fluid par backward pseudo-force act karta hai; oil rear side par higher level aur front par lower level banata hai, free surface inclined hoti hai. *[Q37]*

⚠ **EXAM TRAP — Q36 — Front-Wheel Reaction Depends on Acceleration**

▸ If car accelerates, backward inertial effect causes rearward load transfer, so front-wheel reaction decreases below R. *[Q36]*

▸ If car simply moves at constant velocity on a level road, no longitudinal load transfer occurs and reaction can remain R. Source explanation acceleration interpretation use karti hai. *[Q36]*

15. Average Speed & Acceleration Equations — Core Kinematics Formulae

- Equal-distance halves ko v_1 aur v_2 speeds se cover karne par average speed arithmetic mean nahi hoti: $v_{avg} = 2v_1v_2/(v_1+v_2)$. *[Q40]*
- First equation of motion $v = u + at$ se acceleration $a = (v - u)/t$. *[Q41]*

16. Centrifugation — Washing Machine and Cream Separation

- Washing machine spin cycle centrifugation principle par kaam karta hai; rotation se water/clothes separation facilitate hoti hai. *[Q42]*
- Milk/cream separation exam context mein centrifugal-force based separation se associate ki jati hai: denser phase outward aur less-dense cream relatively inward collect hoti hai. *[Q43]*